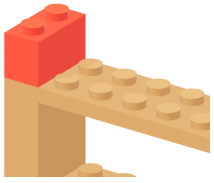
Six cases that express the benchmark goal

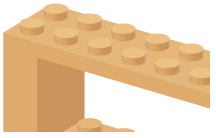
A continuing agent must compare, act legally, update its state, replan, and verify completion.

A Compare and change only the wrong part

PARTIAL / current static API



Wrong-color column



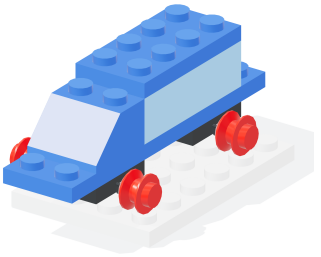
Model: wrong part removed

GPT-6.1 Sol selected and removed the faulty part.
No collateral change; the service stopped before replacement.
This is correct partial action, not completed repair.

B Build a complete object legally

COMPLETED / archived Arena v2

Empty studded baseplate

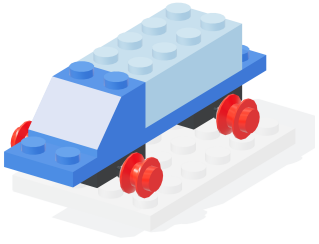


Model: complete car

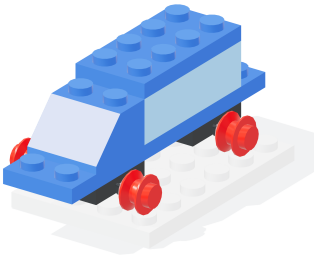
Claude Sonnet 5.5 completed all six placements.
17 decisions: 6 moves, 8 observations, 2 invalid requests.
A legal alternative order can succeed.

C Recover after an unannounced change

COMPLETED / archived Arena v2



A built part was removed

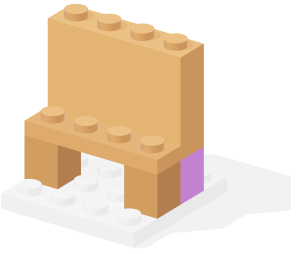


Model: restored and finished

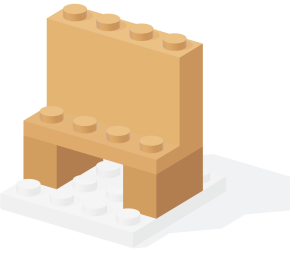
The car run requested fresh evidence and restored the part.
Its normal control succeeded independently.
Recovery is demonstrated; detection time is not known.

D Replan access and verify the result

NOT COMPLETED / archived Arena v2



Model: wrong explicit finish

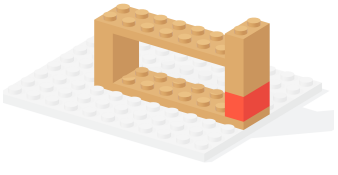


Feasible GT endpoint

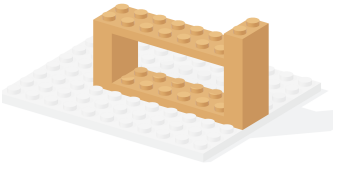
GPT-6.1 Sol built further and encountered blocked access.
Final state: 5 correct, 1 excess, 2 missing placements.
The trace does not identify one internal cognitive cause.

E Open, repair, and restore the context

MODEL UNTESTED / reference only



A buried wrong part

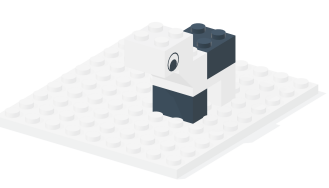


Reference: repaired stage

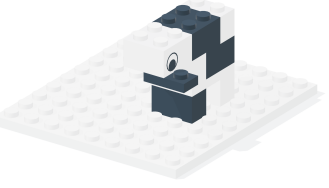
The bookshelf requires legal access work and restoration.
A feasible eight-move witness exists.
No completed API run supports this capability yet.

F Stay correct across repeated changes

MODEL UNTESTED / reference only



Event 1: missing part



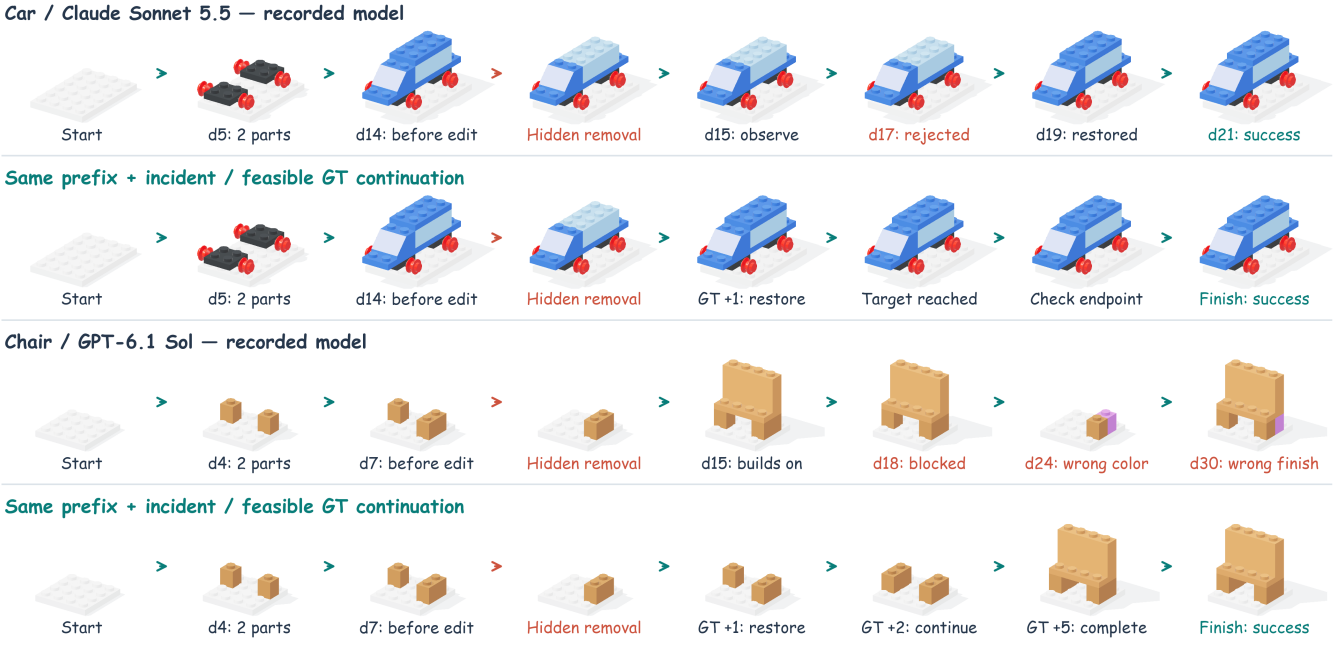
Event 2: displaced part

The cow keeps one task and encounters two separate edits.
The scripted witness recovers and finishes.
This establishes solvability, not learned-agent resilience.

Selected cases, not aggregate scores. High-resolution spectator renders; A is cropped for the reader only.

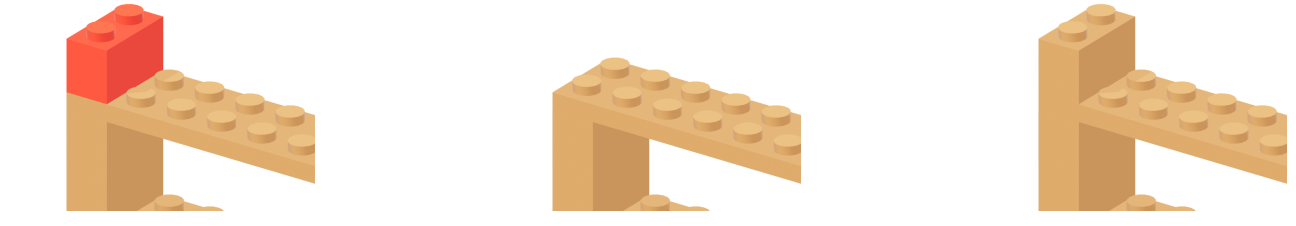
What the recorded models did

Normal success, successful recovery, failed recovery, and technical interruption remain separate.



Top four rows: exact archived v2 model states and feasible, privileged GT continuations.

Current static example: the model removed the correct wrong-color part



Input: wrong-color column | Actual: detach A011 | Required target / reference

Actual calls: 1 inspect A011; 2 detach A011 (accepted); 3 receipt error, no scored action.

13 initially correct placements preserved; the correct replacement is still missing.

Completion is N/A. Confirmed new cost: $0.256830; unresolved reservation: $0.782538.

No extra paid calls after the stop. The old $3 pilot ceiling remains in force; this is not a model failure.

What remains unproven, and how ready the benchmark is

Reference sequences below use private state. Neither row is a model-generated solution.

E Legal access: remove covers, replace the faulty part, restore the stage



Faulty start | Open 1 | Open 2 | Fault removed | Replace | Restore | Target | Finish

F Repeated recovery: one continuous cow episode, two separate incidents



Start | Before edit 1 | Missing | Restored | Before edit 2 | Displaced | Restored | Finish

Selected reference states. Access work is legitimate; references are feasible, not proved shortest.

Readiness review: about 70-75 / 100, excluding large model experiments

Internal project judgment, not a CVPR rating or an acceptance probability.



Before calling the benchmark complete:

1 Match sensing / state / budgets before claiming static-to-dynamic transfer.
2 Finish input adapters and payload checks; reduce avoidable image burden.
3 Audit annotation visibility, action legality, source rights, and evaluation separation.

CVPR basis: technical soundness, contribution, reproducibility, and data release. See the linked review notes.

2027 submission format and LLM disclosure are separate requirements; the current 10-page paper is a working draft.